

Introduction to Cryptography

Spring 2025

Cryptology

- Has two branches
 - Cryptography
 - Cryptanalysis
- Cryptography
 - The science of secret writing with the goal of hiding the meaning of a message.
- Cryptanalysis
 - The science and art of **breaking** cryptosystems.

Cryptanalysis

- Important, because it is the only way to show and understand that a cryptosystem is **secure**.
 - It is not done by organized crime etc. Usually done by academics.
 - In cryptography, breaking a system is *encouraged*.

Cryptography

- Cryptography provides the following:
- **Confidentiality**
 - Unauthorized parties cannot read the message.
- **Integrity**
 - Unauthorized modification can be detected.
- **Authentication**
 - Who sent this message?
- **Non-repudiation**
 - In some systems, the sender cannot plausibly deny an action.
- For now we are mainly studying **confidentiality**.

Basic notation

- m = plaintext / message
- c = ciphertext
- k = key
- Encryption:
 - $c = \text{Enc}(k, m)$
- Decryption
 - $m = \text{Dec}(k, c)$
- $\text{Dec}(k, \text{Enc}(k, m)) = m$

Basics

- Alice, Bob, Eve and Mallory.
- Alice wants to send a message m to Bob.
 - m is called the **plaintext**
- If Alice can transform the message m to c , it is called a **ciphertext**.
- Turning M to C is called **encryption**
- Turning C to M is called **decryption**.
- The aim is to make sure that if someone (Eve maybe) intercepts the communication and read (hear) c , it would be meaningless to them.
 - We encrypt messages so that it is only meaningful to parties which can *decrypt* them.

Basics

- If Bob can understand the message but not Eve, that means Bob has some knowledge that Eve does not have.
 - One can say that we can have a *secret algorithm*. And this was the case for some years. But this is hard to do. If there are a lot of parties, you need to invent a lot of secret algorithms.
- A. Kerckhoff thought about this problem and formulated a set of cryptographic design principles.
 - The method must not be required to be secret, and it must be able to fall into enemy's hands without causing inconvenience.
- In short, we shall design cryptographic systems thinking that an attacker will know everything about the system.

Kerckhoff's Principle

- We always assume the attacker knows:
 - The algorithm
 - The protocol
 - The implementation approach
 - How the system works
- The attacker should still fail because:
 - **They do not know the secret key**
- **Security through obscurity is not a substitute for cryptographic security.**

Basics

- How are we going to provide secrecy?
 - By using **secret keys**
- **Key** is an extra information and a parameter to *encryption* and *decryption* algorithms.
 - So, instead of *hiding* the algorithm, we *hide* the **key**.
- If the key gets compromised, you just create a **new** key instead of creating a new algorithm.
- However, it is not that easy → keygen (key generation)
 - When you combine encryption, decryption and keygen algorithm, you have an **encryption scheme**.

Basics

- The aim is not to hide the communication itself, but the *contents* of it.
 - Trying to hide the **communication** is called **steganography**.
- In addition, cryptography does not care how Alice and Bob are communicating. We only care about the secrecy of c and we'll assume that people are communicating on an **insecure** channel.
- At this point, we are not thinking about Eve changing c . So, it may be possible by using a MITM attack they can intercept the communication and just send gibberish instead.

Encoding

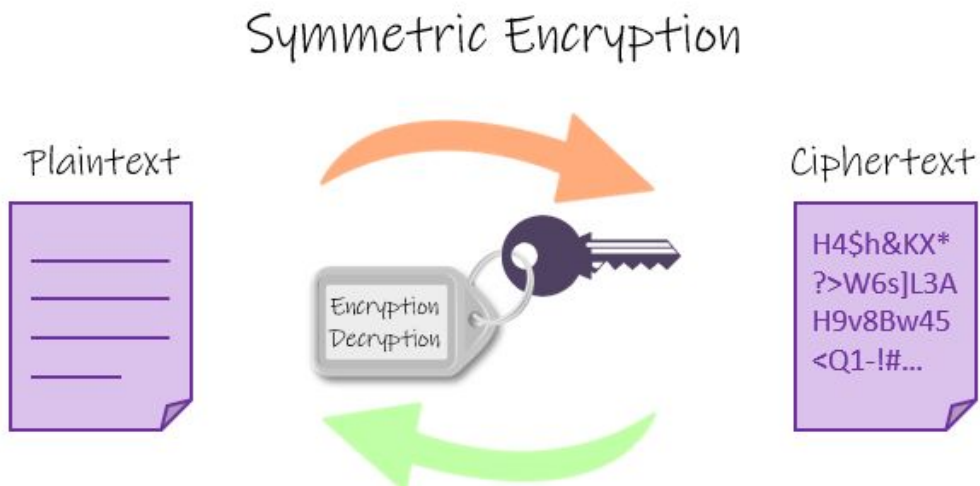
- The process of converting data from one form into another for several reasons
 - Storage, transmission, processing etc.
- Used to ensure that data is represented in a standardized way.
- NOT FOR SECURITY
 - Encoding do not provide any security.
 - Encoding is NOT encryption.
 - In encryption schemes we have a **key**.
- Encoding can be reversed by **decoding**
- Main purpose is representation
 - For example, check out BASE64.
 - Check out ASCII, UTF-8.
 - Check out URL encoding

Steganography

- Means representing information within another message or physical object.
 - When someone looks at it, they cannot understand whether something is hidden or not.
- It is the science and art of covert communication.
 - A captured soldier giving a message by blinking his eyes in Morse Code on live TV.
 - Some printers integrate the model, serial number and timestamps on each printout by using yellow dots.
 - Not recognizable to the naked eye.
 - Similar to ballistic information.
- “Presidents embargo ruling should have immediate notice. Grave situation affecting international law. Statement foreshadows ruin of many neutrals. Yellow journals unifying national excitement immensely”
 - Check the first chars of every word.
- This can be further improved so that you check every second word, etc.
 - Both parties need to know how it works.

Symmetric-key encryption

- Also called *secret-key* encryption. Up until the creation of RSA, this was the only standard.
 - RSA is *asymmetric key* encryption.
- In this method, the **key** used for encryption and decryption is the same **key**.



Substitution Cipher

- Here, we replace each letter with a different letter.
 - We need a table for that.
- For example, we will create a new order alphabet.
- THIS SENTENCE IS ENCRYPTED

C-D-E-H-I-N-P-R-S-T-Y

X-J-L-A-Z-E-V-K-H-O-M

Shift cipher

- It is a *type* of substitution cipher.
 - Caesar cipher is an example to it.
- Each letter in the alphabet represents a number.
 - A:0,
 - B:1
 - ..
 - Z: 25

Key Space

- The key space is the set of all possible keys.
- If there are **N** possible keys:
 - Attacker can try:
 - $k_1, k_2, \dots, k_N$
- For example the *Cesar cipher* we'll see has 26 possible keys, whereas an encryption standard AES has 2^{218} possible keys.
 - You can just try every key in the **Cesar Cipher**!

Caesar Cipher

- Here, we shift all the letters in a text by a certain number of places.
- The **key** for this cipher is a *letter* which represents the number of places for the shift.
 - If the key is D, that means shift 3 keys
 - A means do not shift, Z means -1
- So, word CAESAR becomes RPTHGP with key **P**.
- Easy to create but also easy to crack.
 - Frequency analysis
 - As you can see there are 2 A's in CAESAR and there are two P's in the ciphertext.
 - By checking the most frequent letters in English, this can be solved.

Vigenere Cipher

- Here, the **key** is a *word* instead of a *letter*.
- Let's say our key is **KEY**, and we want to encrypt the word *CRYPTOGRAPHY*
 - As one can see, the length of the key is smaller than the plaintext.
 - So we need to repeat it as needed:
 - CRYPTOGRAPHY
 - KEYKEYKEYKEY
- Then, we use each letter of the key as a shift for the Caesar cipher. So, to encrypt C, we use K (it means 10) as the shift and we get M. We continue like that.
- If the message is shorter than the key, it is a *one time pad*.

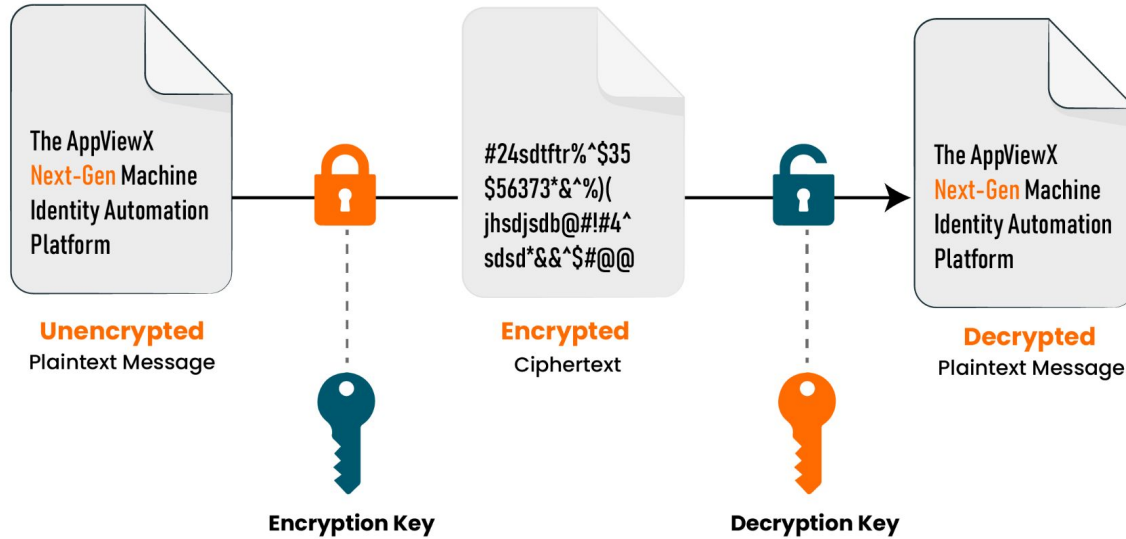
Why classical ciphers fail?

- What went wrong?
- Caesar
 - Tiny key space
 - Preserves language statistics and prone to frequency attacks
- Substitution Cipher
 - Larger key space
 - Still preserves statistical patterns
- Vigenere Cipher
 - Hides simple letter frequencies better
 - Repeated key creates patterns

One-time Pad (Vernam's cipher)

- One of the most important creations in cryptography.
- Remember vigenere cipher?
 - The basic idea here is that making the *key* as long as the *message*. So that it **never** repeats.
 - That repeat is a way to crack it.
 - If there are no repeats, you cannot use frequency analysis.
- In One Time Pad, the key **must be random string**.

Asymmetric Encryption



Asymmetric Encryption

- Unlike **symmetric encryption** where we have a **single key** used for both encryption and decryption, here we have a **key pair**.
- Key pair: public key and private key
 - There is a mathematical relationship between them.
- One is used for **encryption** and the other is used for **decryption**
- We'll see more about this later.
- Examples: RSA, ECDSA, etc.

Other cryptographic primitives

- Hash functions
- MACs: Message Authentication Codes
- Digital Signatures